Digital Logic System

Assignment 1

1. Make 3 question for each of the following and solve them. At least 2 question must be floating point out of those 3.
   a. Binary to decimal
   b. Decimal to binary
   c. Octal to decimal
   d. Decimal to octal
   e. Hexadecimal to decimal
   f. Decimal to hexadecimal
   g. Binary to octal
   h. Octal to binary
   i. Binary to Hexadecimal
   j. Hexadecimal to binary
   k. Octal to hexadecimal
   l. Hexadecimal to octal
2. Subtract using 1's, 2's, 9's and 10's complement. Make 3 question for each of them and solve them.
3. Convert the following:
   $(154)_{BCD} = (?)_{Gray}$
   $(128)_{BCD} = (?)_{Gray}$
   $(457)_{BCD} = (?)_{Gray}$
   $(101110110)_2 = (?)_{Gray}$
4. What is parity bit? How does it help to detect error?
5. Write short note on excess-3 code.
6. Differentiate between analog and digital system with their advantages and disadvantages.